MIL-E-5589C
20 August 1973
SUPERSEDING
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# MILITARY SPECIFICATION

## EARPHONE (EXTENDED RANGE) H-301/U

This specification is approved for use by all Departments and Agencies of the Department of Defense.

## 1. SCOPE

1.1 Scope. This specification covers a dynamic earphone having an impedance of 300 ohms with extended frequency and a flat response (see 6.1).

## 2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of th specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

| | | |
|---|---|---|
| NN-P-71 | - | Pallets, Material Handling, Wood, Double Faced, Stringer Construction. |
| QQ-S-781 | - | Strapping, Steel, Flat and Seals. |
| PPP-B-566 | - | Boxes, Folding, Paperboard. |
| PPP-B-585 | - | Boxes, Wood, Wirebound. |
| PPP-B-601 | - | Boxes, Wood, Cleated - Plywood. |
| PPP-B-621 | - | Boxes, Wood, Nailed and Lockcorner. |
| PPP-B-636 | - | Boxes, Shipping, Fiberboard. |
| PPP-B-676 | - | Boxes, Setup. |
| PPP-T-60 | - | Tape: Pressure-Sensitive Adhesive, Waterproof, For Packaging. |
| PPP-T-76 | - | Tape, Pressure-Sensitive Adhesive Paper, (For Carton Sealing). |

MILITARY

| | | |
|---|---|---|
| MIL-P-116 | - | Preservation, Methods of. |
| MIL-B-43014 | - | Boxes: Water Resistant Paperboard; Folding, Set-up, and Metal-Stayed. |
| MIL-C-45662 | - | Calibration System Requirements. |

STANDARDS

MILITARY

| | | |
|---|---|---|
| MIL-STD-105 | - | Sampling Procedures and Tables for Inspection by Attributes. |
| MIL-STD-129 | - | Marking for Shipment and Storage. |
| MIL-STD-130 | - | Identification Marking of US Military Property. |
| MIL-STD-147 | - | Palletized Unit Loads on 40" x 48" Pallets. |
| MIL-STD-202 | - | Test Methods for Electronic and Electrical Component Parts. |
| MIL-STD-454 | - | Standard General Requirements for Electronic Equipment. |

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.



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American National Standards Institute, Inc

S1.12-1967 - Laboratory Standard Microphones, Specifications for.
Z24.9-1949 - Coupler Calibration of Earphones, Method for the

(Application for copies should be addressed to the American National Standards Institute, Inc, 1430 Broadway, New York, New York 10018.)

(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

3. REQUIREMENTS

3.1 Qualification. Earphones furnished under this specification shall be products which are qualified for listing on the applicable qualified products list at the time set for opening of bids (see 4.4 and 6.3).

3.2 Material. Materials shall be used which will enable the earphones to meet the performance requirements of this specification. Acceptance or approval of any constituent material shall not be construed as a guaranty of the acceptance of the finished product.

3.2.1 Case. Case material shall be of steel or other suitable magnetic shielding material, capable of withstanding headband yoke pressure. The earphone shall have a black finish.

3.2.2 Insulating and impregnating compounds. All insulating and impregnating compounds applied, including varnishes, waxes and the like, shall be suitable for each particular application. Compounds shall adequately preserve the dielectric strength of the insulation to which it is applied. The compound used shall not (either in the state of its original application or as a result of aging) show any injurious effect upon the insulation it is to protect, and shall not cause corrosion or deterioration of adjacent metal or plastic parts.

3.2.3 Magnetic stability. The materials used in the magnetic circuit shall be processed and assembled so that the earphone element will suffer no degradation in performance over long periods of use or storage.

3.2.4 Cementing compounds. Any cementing compound used in the construction of the earphone shall show no evidence of loosening nor have other detrimental effects.

3.2.5 Fungus resistant. Unless certification is furnished, materials for fabrication of earphones covered by this specification shall conform to requirement 4 of MIL-STD-454. When nutrient materials are used, they shall be tested prior to fabrication of the earphones.

3.2.6 Soldering. The materials used for soldering shall conform to requirement 5 of MIL-STD-454.

3.3 Design. Earphones shall be designed in accordance with figure 1. Earphones shall be inherently stable mechanically and acoustically.

3.3.1 Diaphragm. The earphone shall be constructed to have equalization of pressure on both sides of the diaphragm when subjected to changes in ambient pressure as low as 3.4 inches of mercury.

3.3.2 Weight. The weight of each earphone shall not be greater than 3.5 ounces.

3.3.2.1 Transformers. The use of a transformer in the earphone is permissible; however, when a transformer is used it shall be inside the earphone case and included as part of the total weight.

3.3.3 Cap. The earphone cap shall be nonremovable.



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3.3.4 Stay-cord anchor. At the option of the manufacturer, the earphone may be provided with a nonferrous metal eye to permit anchoring of the tie or stay cord of the cordage, as shown in figure 1.

3.4 Performance characteristics.

3.4.1 Frequency response. When measured as specified in 4.6.2, the frequency response at 1,000 hertz (Hz) shall be between 106 and 112 decibels (dB) above a reference level of 0.0002 dyne per square centimeter with 1 milliwatt root mean square (rms) power. The frequency response shall not deviate from the 1,000 Hz response by more than the values shown in table I.

TABLE I. Frequency response deviations.

| Frequency (in Hz) | Deviations (in dB) | |
|---|---|---|
| | Min | Max |
| 100 - 2500 | -3 | +3 |
| 2501 - 3000 | -3 | +5 |
| 3001 - 7000 | -7 | +6 |

3.4.2 Impedance. When measured as specified in 4.6.3, the impedance of the earphone element at 1,000 Hz shall be 300 ±30 ohms.

3.4.3 Harmonic distortion. When measured as specified in 4.6.4, the rms harmonic distortion in the acoustic output of the earphone shall not exceed 5 percent at 200 Hz and shall not exceed 4 percent at any fundamental frequency between 400 and 7,000 Hz.

3.4.4 Dielectric withstanding voltage. When tested as specified in 4.6.5, there shall be no arcing or breakdown of the insulation.

3.4.5 Stray magnetic field. When measured as specified in 4.6.6, the deflection of the magnetic compass shall not be greater than 7 1/2 degrees.

3.4.6 Endurance. When tested as specified in 4.6.7, there shall be no audible rattles or other signs of failure, and the frequency response shall not vary more than ±3 dB from the initial reading recorded in 3.4.1.

3.4.7 Moisture resistance. When tested as specified in 4.6.8, the frequency response, harmonic distortion, and dielectric withstanding voltage shall be as specified in 3.4.1, 3.4.3 and 3.4.4 respectively. There shall be no loosening or deformation of parts or other damage.

3.4.8 Barometric pressure (reduced). When earphones are tested as specified in 4.6.9, the frequency response shall not vary more than ±3 dB from the initial reading recorded in accordance with 3.4.1.

3.4.9 Thermal shock. When earphones are tested as specified in 4.6.10, the frequency response shall not vary more than ±3 dB from the initial reading recorded in accordance with 3.4.1.

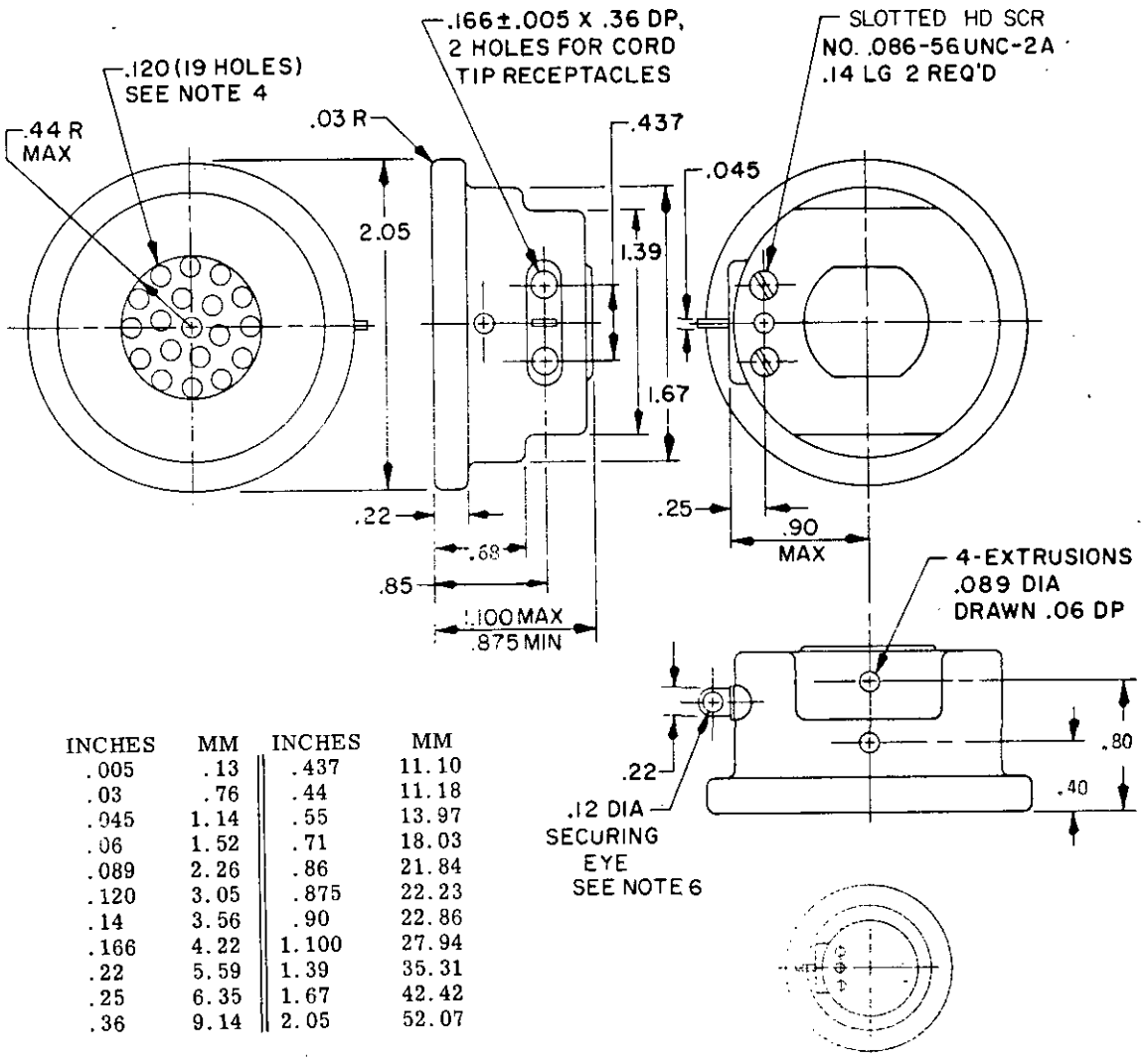
3.4.10 Vibration, high frequency. When earphones are tested as specified in 4.6.11, there shall be no structural failure or other defects, and the frequency response shall not vary more than ±3 dB from the initial reading recorded in accordance with 3.4.1.

3.4.11 Shock. The earphones shall exhibit no damage or failure from breaking, cracks, deformation or loosening of parts when tested as specified in 4.6.12. The frequency response of the earphone shall not vary more than ±3 dB from the initial reading recorded in accordance with 3.4.1.

3.4.12 Salt spray (corrosion). After exposure the earphones shall be tested as specified in 4.6.13 (c). Within 3 hours after the drying period, the earphones shall be tested as specified in 4.6.13 (e). There shall be no evidence of corrosion or pitting which in anyway interfers with electrical or mechanical performance.



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| INCHES | MM | INCHES | MM |
|---|---|---|---|
| .005 | .13 | .437 | 11.10 |
| .03 | .76 | .44 | 11.18 |
| .045 | 1.14 | .55 | 13.97 |
| .06 | 1.52 | .71 | 18.03 |
| .089 | 2.26 | .86 | 21.84 |
| .120 | 3.05 | .875 | 22.23 |
| .14 | 3.56 | .90 | 22.86 |
| .166 | 4.22 | 1.100 | 27.94 |
| .22 | 5.59 | 1.39 | 35.31 |
| .25 | 6.35 | 1.67 | 42.42 |
| .36 | 9.14 | 2.05 | 52.07 |



OPTIONAL FORM VIEW
SEE NOTE 5

NOTES:
1. Dimensions are in inches.
2. Metric equvalents (to the nearest .01 mm) are given for general information only and are based upon 1 inch = 25.4 mm.
3. Unless otherwise specified, tolerances are ± .02 (.51 mm) for two-place decimals, ± .010 (.25 mm) for three-place decimals, and ± 1/2 degree on angles.
4. Size, number and location of holes in cap to be determined by manufacturer, subject to accoustical and electrical requirements and limiting dimensions. A single comparatively large hole is not acceptable.
5. The form factor may conform to the optional view shown. However, it must conform with the above dimensions.
6. The stay cord anchor may be provided at the option of the manufacturer.

FIGURE 1. Earphone.



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3.4.13 Acoustic quality. When earphones are tested as specified in 4.6.14, there shall be no buzzing, rattling, nor other spurious sounds in the acoustic output of the earphone.

3.5 Marking. Earphones shall be marked in accordance with MIL-STD-130 with the type designation and manufacturer's code symbol or name.

3.6 Workmanship. Earphones shall be processed in such a manner as to be uniform in quality and shall be free from loose parts, excess solder, sharp edges, burrs, flaking, scratches, peeling, metal chips and foreign material. Soldering, impregnation of coils, plating, staking, riveting, and machine screw assemblage shall be neat and thorough. The coil shall be centered properly, void of foreign objects in the air gap and shall not drag on the frame or magnet.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Test equipment and inspection facilities. The supplier shall establish and maintain a calibration system in accordance with MIL-C-45662.

4.2 Classification of inspections. The inspections specified herein are classified as follows:

(a) Qualification inspection (see 4.4).
(b) Quality conformance inspection (see 4.5).

4.3 Inspection conditions. Unless otherwise specified herein, all inspections shall be performed in accordance with the test conditions specified in the "GENERAL REQUIREMENTS" of MIL-STD-202.

4.4 Qualification inspection. Qualification inspection shall be performed at a laboratory acceptable to the Government (see 6.3) on sample units produced with equipment and procedures normally used in production.

4.4.1 Sample size. Six earphones shall be subjected to the qualification inspection.

4.4.2 Inspection routine. The sample shall be subjected to the inspections specified in table II, in the order shown. All sample units shall be subjected to the inspections of group I. The sample shall then be divided equally into two groups of three units each, and subjected to the inspection for their particular group.

4.4.3 Failures. One or more failures shall be cause for refusal to grant qualification approval.

4.4.4 Retention of qualification. To retain qualification, the supplier shall forward a report at 12-month intervals to the qualifying activity. The qualifying activity shall establish the initial reporting date. The report shall consist of:

(a) A summary of the results of the tests performed for inspection of product for delivery, group A and B, indicating as a minimum the number of lots that have passed and the number that have failed. The results of tests of all reworked lots shall be identified and accounted for.
(b) A summary of the results of tests performed for qualification verification inspection, group C, indicating as a minimum the number and mode of subgroup failures. The summary shall include results of all qualification verification inspection tests performed on completed lots during the 12-month period. If the summary of the test results indicates nonconformance with specification requirements, and corrective action acceptable to the qualifying activity has not been taken, action may be taken to remove the failing product from the qualified products list.



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Failure to submit the report within 30 days after the end of each 12-month period may result in loss of qualification for the product. In addition to the periodic submission of inspection data, the supplier shall immediately notify the qualifying activity at any time that the inspection data indicates noncompliance of the product to meet the requirements of this specification.

In the event that no product occurred during the reporting period, a report shall be submitted certifying that the company still has the capabilities and facilities necessary to produce the item. If during 2 consecutive reporting periods there has been no production, the manufacturer may be required, at the discretion of the qualifying activity, to submit the earphones to testing in accordance with the qualification inspection requirements.

TABLE II. Qualification inspection.

| Examination or test | Requirement paragraph | Test method paragraph |
|---|---|---|
| Group I | | |
| Visual and mechanical examination | 3.3, 3.5, and 3.6 | 4.6.1 |
| Frequency response | 3.4.1 | 4.6.2 |
| Impedance | 3.4.2 | 4.6.3 |
| Harmonic distortion | 3.4.3 | 4.6.4 |
| Dielectric withstanding voltage | 3.4.4 | 4.6.5 |
| Stray magnetic field | 3.4.5 | 4.6.6 |
| Endurance | 3.4.6 | 4.6.7 |
| Acoustic quality | 3.4.13 | 4.6.14 |
| Group II | | |
| Moisture resistance | 3.4.7 | 4.6.8 |
| Group III | | |
| Barometric pressure (reduced) | 3.4.8 | 4.6.9 |
| Thermal shock | 3.4.9 | 4.6.10 |
| Vibration, high frequency | 3.4.10 | 4.6.11 |
| Shock | 3.4.11 | 4.6.12 |
| Salt spray (corrosion) | 3.4.12 | 4.6.13 |

4.5 Quality conformance inspection.

4.5.1 Inspection of product for delivery. Inspection of product for delivery shall consist of groups A and B inspections. Except as specified in 4.5.2.1.4, delivery of products which have passed the groups A and B inspections shall not be delayed pending the results of the group C inspection.

4.5.1.1 Inspection lot. An inspection lot shall consist of all earphones of the same type produced under essentially the same conditions, and offered for inspection at one time.

4.5.1.2 Group A inspection. Group A inspection shall consist of the examinations and tests specified in table III, in the order shown.

4.5.1.2.1 Sampling plan. Statistical sampling and inspection shall be in accordance with MIL-STD-105 for general inspection level II. The acceptable quality level (AQL) shall be as specified in table III. Major and minor defects shall be as defined in MIL-STD-105.



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TABLE III. Group A inspection.

| Examination or test | Requirement paragraph | Test method paragraph | AQL (percent defective | |
|---|---|---|---|---|
| | | | Major | Minor |
| Visual and mechanical examination | 3.3, 3.5 and 3.6 | 4.6.1 | 1.0 | 4.0 |
| Frequency response | 3.4.1 | 4.6.2 | 1.0 | None |
| Impedance | 3.4.2 | 4.6.3 | | |
| Harmonic distortion | 3.4.3 | 4.6.4 | | |
| Acoustic quality 1/ | 3.4.13 | 4.6.14 | | |

1/ This test is to be performed on a 100 percent basis.

4.5.1.2.2 Rejected lots. If an inspection lot is rejected, the supplier may rework it to correct the defects, or screen out the defective units, and resubmit for reinspection. Resubmitted lots shall be inspected using tightened inspection. Such lots shall be separate from new lots, and shall be clearly identified as reinspected lots.

4.5.1.3 Group B inspection. Group B inspection shall consist of the examinations and tests specified in table IV in the order shown, and shall be made on sample units which have been subjected to and have passed the group A inspection.

4.5.1.3.1 Sampling plan. The sampling plan shall be in accordance with MIL-STD-105 for special inspection level S-4. The sample size shall be based on the inspection lot size from which the sample was selected for group A inspection. The AQL shall be 2.5 percent defective.

TABLE IV. Group B inspection.

| Examination or test | Requirement paragraph | Test method paragraph |
|---|---|---|
| Dielectric withstanding voltage | 3.4.4 | 4.6.5 |
| Stray magnetic field | 3.4.5 | 4.6.6 |

4.5.1.3.2 Rejected lots. If an inspection lot is rejected, the supplier may rework it to correct the defects, or screen out the defective units, and resubmit for reinspection. Resubmitted lots shall be inspected using tightened inspection. Such lots shall be separate from new lots, and shall be clearly identified as reinspected lots.

4.5.1.3.3 Disposition of sample units. Sample units which have passed all the group B inspection may be delivered on the contract or purchase order, if the lot is accepted and the sample units are still within specified electrical tolerances.

4.5.2 Periodic inspection. Periodic inspection shall consist of group C inspection.

4.5.2.1 Group C inspection. Group C inspection shall consist of the examinations and tests specified in table V, in the order shown. Group C inspection shall be made on sample units selected from inspection lots which have passed the groups A and B inspections.

4.5.2.1.1 Sampling plan. Four sample units shall be selected from each two-months' production, or each 1,000 units whichever occurs first, after date of notification of qualification.



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TABLE V. Group C inspection.

| Examination or test | Requirement paragraph | Test method paragraph | Number of sample units to be inspected |
|---|---|---|---|
| Subgroup 1 | | | |
| Endurance | 3.4.6 | 4.6.7 | 2 |
| Subgroup 2 | | | |
| Moisture resistance | 3.4.7 | 4.6.8 | 2 |
| Visual and mechanical examination | 3.3, 3.5, and 3.6 | 4.6.1 | |

4.5.2.1.2 Failures. If one or more sample units fail to pass group C inspection, the sample shall be considered to have failed.

4.5.2.1.3 Disposition of sample units. Sample units which have been subjected to group C inspection shall not be delivered on the contract or purchase order.

4.5.2.1.4 Noncompliance. If a sample fails to pass group C inspection, the supplier shall take corrective action on the materials or processes, or both, as warranted, and on all units of product which can be corrected and which were manufactured under essentially the same conditions, with essentially the same materials, processes, etc., and which are considered subject to the same failure. Acceptance of the product shall be discontinued until corrective action, acceptable to the Government, has been taken. After the corrective action has been taken, group C inspection shall be repeated on additional sample units (all inspection, or the inspection which the original sample failed, at the option of the Government). Groups A and B inspections may be reinstituted; however, final acceptance shall be withheld until the group C reinspection has shown that the corrective action was successful. In the event of failure after reinspection, information concerning the failure and corrective action taken shall be furnished to the cognizant inspection activity and the qualifying activity.

4.5.3 Inspection of preparation for delivery. The sampling and inspection of the preservation-packaging and interior package marking shall be in accordance with the group A and B quality conformance inspection requirements of MIL-P-116. The sampling and inspection of the packing and marking for shipment and storage shall be in accordance with the quality assurance provisions of the applicable container specification and the marking requirements of MIL-STD-129.

4.6 Methods of examination and test.

4.6.1 Visual and mechanical examination. Earphones shall be examined to verify that the design, physical dimensions, weight, marking, and workmanship are in accordance with the applicable requirement (see 3.3, 3.5 and 3.6).

4.6.2 Frequency response (see 3.4.1). The earphone shall be mounted as shown in the test circuit (see figure 2). Adjust the output from the audio oscillator to 1.1 volts rms at 1,000 Hz as measured by the vacuum tube voltmeter (VTVM No. 1). For qualification inspection, the frequency response shall be measured and recorded in accordance with 4.6.2.1. For quality conformance inspection the frequency response shall be measured and recorded in accordance with 4.6.2.1 or 4.6.2.2, at the option of the supplier.

4.6.2.1 Automatic recorder. The acoustic output shall be recorded on a direct writing strip graph using an automatic plotter or curve tracer with a minimum writing speed of 10 inches per second and a maximum chart speed of 30 inches per minute. The output shall be recorded in the dB value above 0.0002 dyne per square centimeter. The writing speed and chart speed shall be noted.



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4.6.2.2 Point-to-point. Measure the acoustic output from the calibrated microphone and preamplifier unit as indicated by the reading of VTVM No. 2 (see figure 2), and convert this reading to the equivalent dB value above 0.0002 dyne per square centimeter, using the most recent available calibration curve for the test microphone. Measure and record the dB increments of 100 Hz from 100 to 1,000 Hz and at increments of 250 Hz from 1,000 to 8,000 Hz. The graph shall show the dB values on the ordinate scale, reading from 80 to 140 dB above 0.0002 dyne per square centimeter, and the Hz value on the abscissa scale, reading from 100 to 10,000 Hz inclusive.

4.6.3 Impedance (see 3.4.2). Impedance measurements shall be made at 1,000 Hz with the earphone mounted on the coupler as shown on figure 2. The 1,000 Hz voltage applied across the terminals of the earphone element during impedance measurements shall be between 0.9 and 1.3 volts rms. The impedance shall be measured by measuring the voltage across, and the current through the earphone.

4.6.3.1 Voltage method. At the option of the manufacturer, the impedance may be calculated by using a voltmeter. Using a circuit similar to that shown in figure 2, measure the voltage across the 300 ohm resistor and across the earphone under test. The earphone impedance (Z) is equal to

$$300 \times \frac{V_B}{V_a - V_b}.$$

4.6.4 Harmonic distortion (see 3.4.3). The earphone shall be tested while connected to the test circuit and mounted on the coupler as shown in figure 2. Distortion shall be measured at 200 Hz and at frequencies between 400 and 7,000 Hz, as are required to determine at what frequencies (in this range) maximum distortion exists.

4.6.4.1 Distortion analyzer. The total harmonic distortion shall be determined by a distortion analyzer having the following minimum requirements:

(a) Frequency range - Fundamental frequency from 20 to 20,000 Hz.
(b) Frequency calibration - ±2 percent from 20 to 20,000 Hz.
(c) Harmonic measurement - ±3 percent of full scale value for distortion levels as low as 0.5 percent.

4.6.5 Dielectric withstanding voltage (see 3.4.4). The earphone shall be tested in accordance with method 301 of MIL-STD-202. The following details shall apply:

(a) Test voltage - 500 volts.
(b) Nature of potential - 60 Hz.
(c) Points of application - Earphone: Between one of the terminals and the clamping ring, the metallic case, or other metallic surface which is not covered by a nonconducting coating. Simultaneously, the coil shall be excited with 9.5 volts rms at 200 Hz.

4.6.6 Stray magnetic field (see 3.4.5). Stray magnetic field measurements shall be made in a shielded room or in an area free from appreciable magnetic disturbances. The earphone shall be rotated on its axis at various points of an 8 inch radius sphere surrounding a Kueffel and Esser type No. 80-0090 compass, or equal.

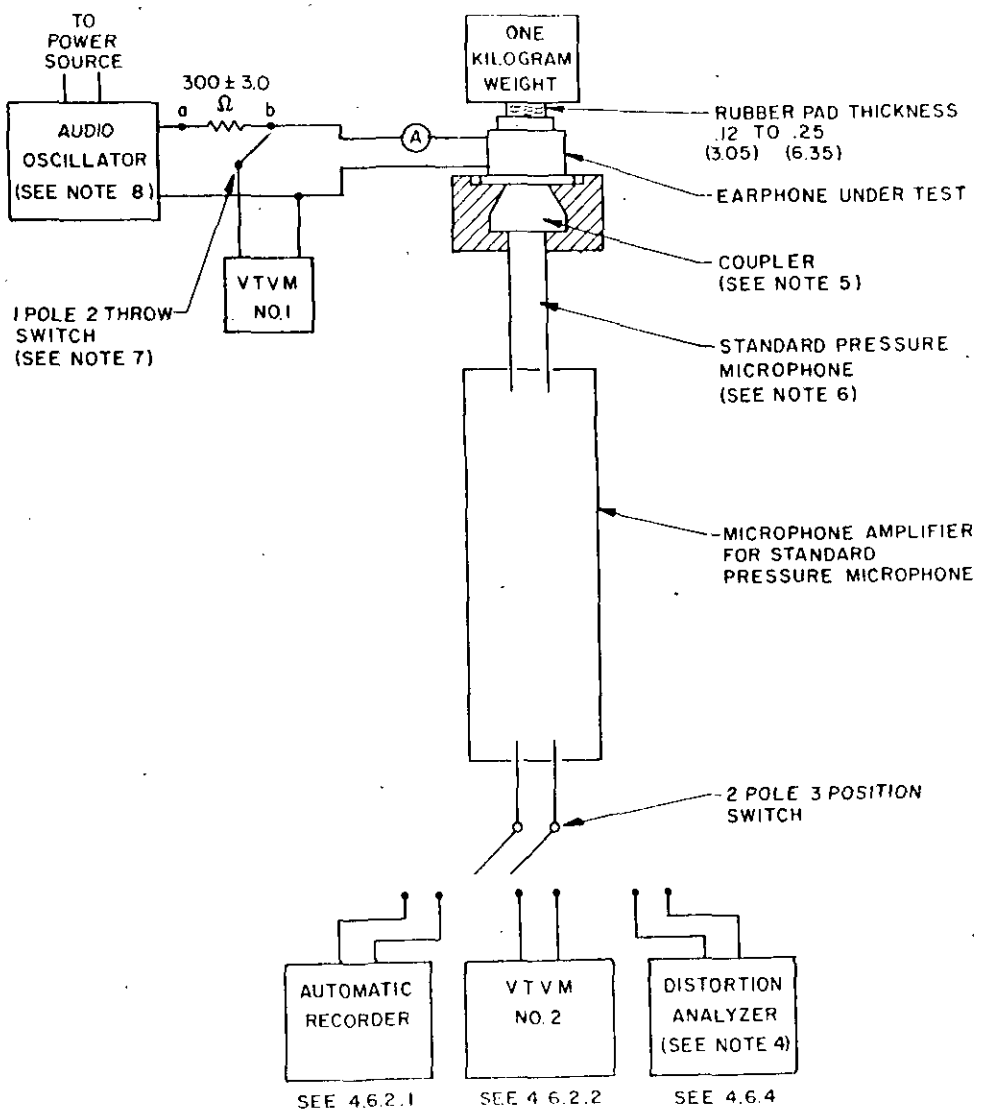
4.6.7 Endurance (see 3.4.6). Audio power of 300 milliwatts rms at 1,000 Hz shall be applied continuously to each earphone under test for 8 hours. During the last hour of test, the earphones shall be checked for audible rattles or other signs of failure. At the end of the 8 hours, the frequency shall be measured as specified in 4.6.2.

4.6.8 Moisture resistance (see 3.4.7). The earphone shall be tested in accordance with method 106 of MIL-STD-202. The following details and exceptions shall apply:

(a) Initial measurements - Not applicable.
(b) Mounting - Any convenient mounting with the front face of the earphone exposed and parallel with the vertical plane.
(c) Loading voltage - Not applicable.



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NOTES:
1. Dimensions are in inches.
2. Metric equivalents (to the nearest .01 mm) are given for general information only and are based upon 1 inch = 25.4 mm.
3. Metric equivalents are in parentheses.
4. See 4.6.4.1 for type and requirement for distortion analyzer.
5. Coupler in accordance with ANSI Z24.9-1949.
6. Standard pressure microphone, in accordance with ANSI S1.12-1967.
7. Use switch position B for paragraph 4.6.3 and switch position A for all other paragraphs requiring electrical tests.
8. Audio oscillator output shall have characteristic output impedance of 600 ohms.
9. "Off" position on switches is optional.
10. Either voltmeter method or voltmeter-ammmeter method is acceptable.

FIGURE 2. Typical test circuit.



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(d) Final measurements - At the completion of the tenth cycle and following a 24-hour period at 25° ±5° C and 50 ±5 percent relative humidity, the earphone element shall be tested for frequency response, harmonic distortion, and dielectric withstanding voltage in accordance with 4.6.2, 4.6.4, and 4.6.5 respectively, and examined for loose or deformed parts and other damage.

4.6.9 Barometric pressure (reduced) (see 3.4.8). Earphones shall be tested in accordance with method 105 of MIL-STD-202. The following details shall apply:

(a) Method of mounting - Not applicable.
(b) Test condition letter - B.
(c) Tests during subjection to reduced pressure - None required.
(d) Tests after subjection to reduced pressure - Frequency response in accordance with 4.6.2.
(e) *Exposure time prior to measurements - 10 minutes.*

4.6.10 Thermal shock (see 3.4.9). Earphones shall be tested in accordance with method 107 of MIL-STD-202. The following details shall apply:

(a) Test condition letter - A.
(b) Measurement after cycling - Frequency response in accordance with 4.6.2.

4.6.11 Vibration, high frequency (see 3.4.10). Earphones shall be tested in accordance with method 204 of MIL-STD-202. The following details shall apply:

(a) Mounting specimens - Diaphragm of the earphone shall be in a vertical plane on the vibrating machine.
(b) Electrical load conditions - 60 milliwatts rms at 1,000 Hz applied to the earphones.
(c) Test condition letter - A.
(d) Measurement after vibration - Frequency response in accordance with 4.6.2.

4.6.12 Shock (specified pulse) (see 3.4.11). Earphones shall be tested in accordance with method 213 of MIL-STD-202. The following details shall apply:

(a) Mounting method - The earphone shall be rigidly attached to an auxiliary mounting plate. The auxiliary mounting plate shall then be rigidly attached to the anvil plate of the shock machine.
(b) Test condition letter - G.
(c) Measurement after shock - Frequency response in accordance with 4.6.2.

4.6.13 Salt spray (corrosion) (see 3.4.12). Earphones shall be tested in accordance with method 101 of MIL-STD-202. The following details and exceptions shall apply:

(a) Applicable salt solution - 5 percent.
(b) Test condition letter - A.
(c) Measurement after exposure - Acoustic quality, in accordance with 4.6.14.
(d) Additional conditioning - Earphones shall be cleaned as prescribed, and dried for 48 hours in a chamber having an ambient temperature of 16 to 32° C and a maximum relative humidity of 30 percent.
(e) Additional measurement - Within 3 hours after the drying period, the earphones shall be tested in accordance with 4.6.2.

4.6.14 Acoustic quality (see 3.4.13). The 1.1 ±0.1 volt rms shall be applied continuously to the terminals of the earphone. The frequency shall be varied continuously from 100 to 7,000 Hz and back to 100 Hz. The acoustic output shall be checked for buzzing, rattling, and other spurious sounds.



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## 5. PREPARATION FOR DELIVERY

(The preparation for delivery requirements specified herein apply only for direct Government procurements. Preparation for delivery requirements of referenced documents listed in section 2 do not apply unless specifically stated in the contract or order. Preparation for delivery requirements for products procured by contractors shall be specified in the individual orders.)

5.1 Preservation-packaging. Preservation-packaging shall be level A or C, as specified (see 6.2).

5.1.1 Level A.

5.1.1.1 Cleaning. Earphones shall be cleaned in accordance with MIL-P-116, process C-1.

5.1.1.2 Drying. Earphones shall be dried in accordance with MIL-P-116.

5.1.1.3 Preservative application. Preservatives shall not be used.

5.1.1.4 Unit packaging. Unless otherwise specified (see 6.2), earphones shall be individually packaged in accordance with MIL-P-116, submethod IA-8 insuring compliance with the general requirements paragraph under methods of preservation (unit protection) and the physical protection requirements paragraph therein.

5.1.1.5 Intermediate packaging. Earphones, packaged as specified in 5.1.1.4, shall be placed in intermediate containers conforming to PPP-B-566 or PPP-B-676. Intermediate containers shall be uniform in size, shape, and quantities, shall be of minimum tare and cube, and shall contain multiples of five unit packages not to exceed 50 unit packages. No intermediate packaging is required when the total quantity shipped to a single destination is less than 50 unit packages.

5.1.2 Level C. Earphones shall be clean, dry and packaged in a manner that will afford adequate protection against corrosion, deterioration and physical damage during shipment from supply source to the first receiving activity.

5.2 Packing. Packing shall be level A, B, or C, as specified (see 6.2).

5.2.1 Level A. The packaged earphones shall be packed in fiberboard containers conforming to PPP-B-636, class weather resistant, style optional, special requirements. In lieu of the closure and waterproofing requirement in the appendix of PPP-B-636, closure and waterproofing shall be accomplished by sealing all seams, corners and manufacturer's joint with tape, 2 inches minimum width, conforming to PPP-T-60, class 1 or PPP-T-76. Banding (reinforcement requirements) shall be applied in accordance with the appendix to PPP-B-636 using nonmetallic or tape banding only.

5.2.2 Level B. The packaged earphones shall be packed in fiberboard containers conforming to PPP-B-636, class domestic, style optional, special requirements. Closures shall be in accordance with the appendix thereto.

5.2.3 Level C. The packaged earphones shall be packed in shipping containers in a manner that will afford adequate protection against damage during direct shipment from the supply source to the first receiving activity. These packs shall conform to the applicable carrier rules and regulations.

5.2.4 Unitized loads. Unitized loads, commensurate with the level of packing specified in the contract or order, shall be used whenever total quantities for shipment to one destination equal 40 cubic feet or more. Quantities less than 40 cubic feet need not be unitized. Unitized loads shall be uniform in size and quantities to the greatest extent practicable.

5.2.4.1 Level A. Earphones, packed as specified in 5.2.1, shall be unitized on pallets in conformance with MIL-STD-147, load type I, with a fiberboard cap (storage aid 4) positioned over the load.



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5.2.4.2 Level B. Earphones, packed as specified in 5.2.2, shall be unitized as specified in 5.2.4.1 except that the fiberboard caps shall be class domestic.

5.2.4.3 Level C. Earphones, packed as specified in 5.2.3, shall be unitized with pallets and caps of the type, size and kind commonly used for the purpose and shall conform to the applicable carrier rules and regulations.

5.3 Marking. In addition to any special marking required by the contract or purchase order (see 6.2), each unit package, intermediate and exterior container, and unitized load shall be marked in accordance with MIL-STD-129.

5.4 General.

5.4.1 Exterior containers. Exterior containers (see 5.2.1, 5.2.2, and 5.2.3) shall be of a minimum tare and cube consistant with the protection required and shall contain equal quantities of identical stock numbered items to the greatest extent practicable.

5.4.2 Army procurements.

5.4.2.1 Level A unit and intermediate packaging. MIL-P-116 submethod IC-1 shall be used in lieu of submethod IA-8. All intermediate containers shall be either weather (or water) resistant or overwrapped with waterproof barrier materials. Containers conforming to PPP-B-566 or PPP-B-676 shall be overwrapped with waterproof barrier materials or shall conform to MIL-B-43014 (see 5.1.1.4 and 5.1.1.5).

5.4.2.2 Level A and B packing. For level A packing when quantities per destination are less than a unitized load, the fiberboard containers shall not be banded but shall be placed in a close fitting box conforming to PPP-B-601, overseas type; PPP-B-621, class 2 style 4 or PPP-B-585, class 3, style 2 or 3. Closure and strapping shall be in accordance with applicable container specification except that metal strapping shall conform to QQ-S-781, type I, class B. When the gross weight exceeds 200 pounds or the container length and width is 48 x 24 inches or more and the weight exceeds 100 pounds, 3 x 4 inch skids (laid flat) shall be applied in accordance with the requirements of the container specification. If not described in the container specification, the skids shall be applied in a manner which will adequately support the item and facilitate the use of material handling equipment. For level B packing, fiberboard boxes shall be weather resistant as specified in level A, and the containers shall be banded (see 5.2.1 and 5.2.2).

5.4.2.3 Level A and B unitization. For level A and B unitization, the fiberboard caps shall be weather resistant and softwood pallets conforming to NN-P-71, type IV, size 2 shall be used (see 5.2.4.1 and 5.2.4.2).

5.4.3 Air Force requirements. For Air Force requirements submethod IC-3 shall be used in lieu of submethod IA-8 (see 5.1.1.4).

6. NOTES

6.1 Intended use. Earphones covered by this specification are used in extended range headsets including the H-216/U.

6.2 Ordering data. *Procurement documents should specify the following:*

(a) Title, number and date of this specification.

(b) Levels of preservation-packaging and packing required (see 5.1 and 5.2).

(c) Method of preservation, if other than submethod IA-8 (see 5.1.1.4, 5.4.2.1, and 5.4.3).

(d) Special marking, if required (see 5.3).



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6.3 Qualification. With respect to products requiring qualification, awards will be made only for products which are at the time set for opening of bids, qualified for inclusion in the applicable qualified products list whether or not such products have actually been so listed by that date. The attention of the suppliers is called to this requirement and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. The activity responsible for the qualified products list is the Naval Electronic Systems Command, ELEX 5043, Department of the Navy, Washington, D.C. 20360. Information pertaining to qualification of products may be obtained from the Naval Electronic Systems Command or the Defense Electronics Supply Center (DESC-EQ), Dayton, Ohio 45444, agent for administration of the qualified products list.

6.4 Changes from previous issue. The extent of changes (deletions, additions, etc.) preclude the annotation of the individual changes from the previous issue of this document.

Custodians:
Army - EL
Navy - EC
Air Force - 80

Review activities:
Army - MI
Navy -
Air Force -
DSA - ES

User activities:
Army - MD
Navy - OS, AS
Air Force - None

Preparing activity:
Navy - EC

Agent:
DSA - ES

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